Supplemental Appendix S1. Detailed Search Strategies

EMBASE 4231 articles:

('heart left bundle branch block'/exp OR (bundle AND left) OR lbb*) AND ('heart pacing'/exp OR 'heart pacing' OR 'pacing'/exp OR pacing) AND ('randomized controlled trial'/exp OR 'controlled clinical trial'/exp OR randomized:ti,ab OR placebo:ti,ab OR 'drug therapy':lnk OR randomly:ti,ab OR trial:ti,ab OR groups:ti,ab OR 'clinical article'/exp OR 'controlled study'/exp OR 'major clinical study'/exp OR 'prospective study'/exp OR 'cohort analysis'/exp OR 'cohort':ti,ab OR 'compared':ti,ab OR 'groups':ti,ab OR 'case control':ti,ab OR 'multivariate':ti,ab OR 'retrospective study' OR prospective*:ti,ab OR retrospective*:ti,ab) AND [english]/lim

PubMed: 1957 articles:

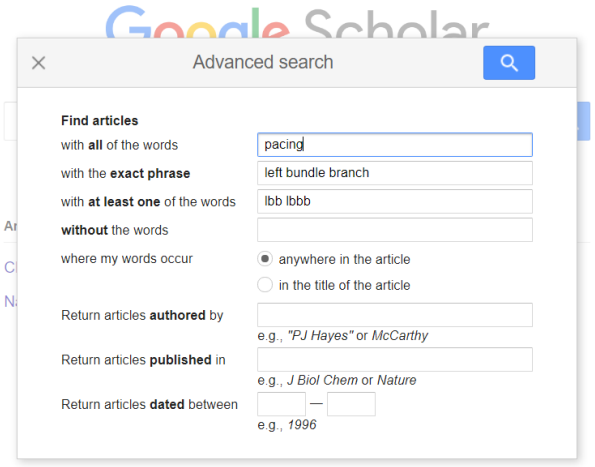
Search: ((("cohort studies"[mesh] OR "case-control studies"[mesh] OR "comparative study"[pt] OR "risk factors"[mesh] OR "cohort"[tw] OR "compared"[tw] OR "groups"[tw] OR "case control"[tw] OR "multivariate"[tw]) OR ((clinical[tiab] AND trial[tiab]) OR "clinical trials as topic"[mesh] OR "clinical trial"[pt] OR random*[tiab] OR "random allocation"[mesh] OR "therapeutic use"[sh])) OR (retrospective study) OR prospective*[tiab] OR retrospective*[tiab]) AND ((("cardiac pacing, artificial"[MeSH Terms]) OR (pacing)) AND (((bundle AND left) OR lbb OR lbbb) OR (left bundle branch block))) Filters: English

Web of Science Gray literature search: Results: 183 (from Web of Science Core Collection) You searched for: TOPIC: (('heart left bundle branch block' OR (bundle AND left) OR lbb*)) AND TOPIC: (pacing) Refined by: DOCUMENT TYPES: (MEETING ABSTRACT OR PROCEEDINGS PAPER) Timespan: All years. Indexes: SCI-EXPANDED, SSCI, A&HCI, ESCI.

Cochrane clinical trials search, 233 results:

(('heart left bundle branch block' OR ((bundle AND left) OR lbb*))) AND (pacing):ti,ab,kw

Google scholar first 56 articles:



Supplemental Table S1. Quality assessment of the four studies with comparative groups using the New-Castle Ottawa Quality Assessment Scale.[15] A maximum of one star can be awarded for each category. A maximum of two stars can be awarded for the comparability column. A score of ≥7 is considered to be of good quality.

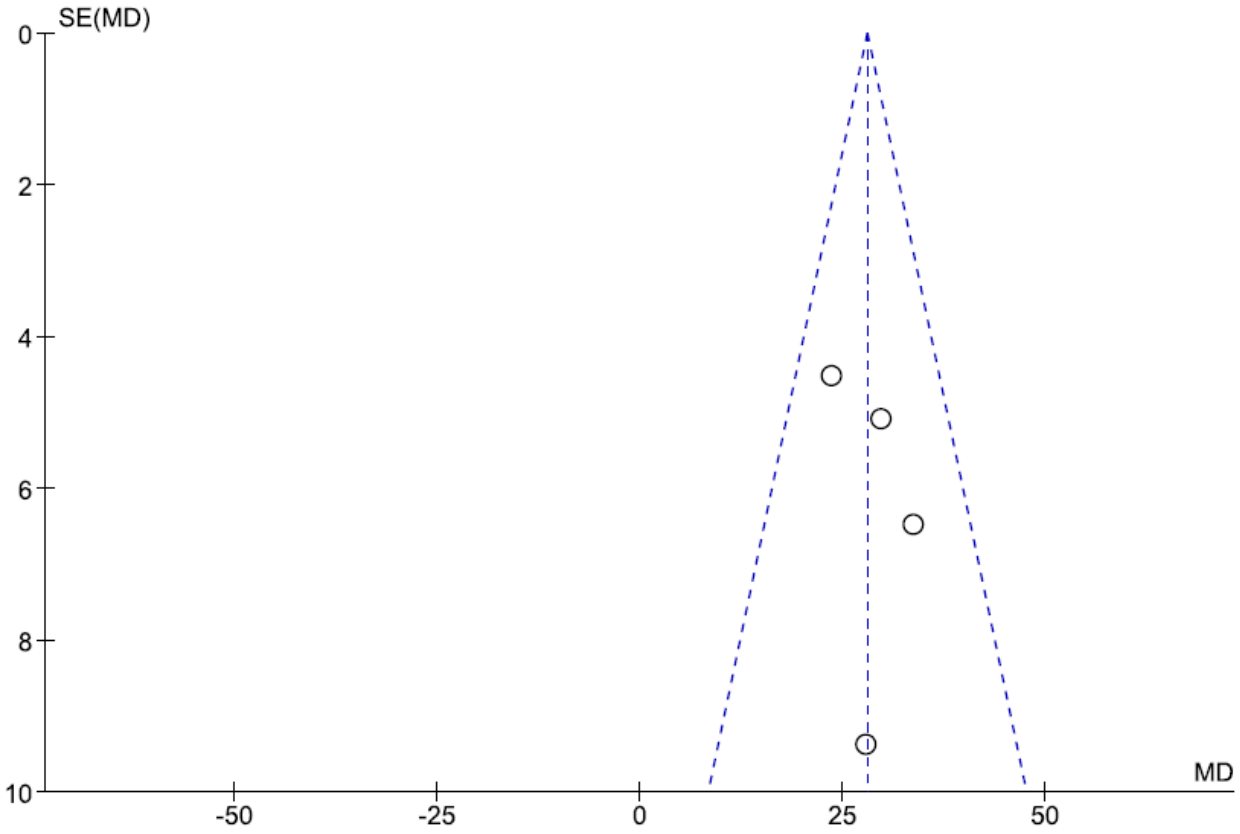
Study, Year (Ref #)	Wang et al., 2020[14]	Li et al., 2020[13]	Wu et al., 2020[9]	Guo et al., 2020
Representative of the exposed cohort	*	*	*	*
Selection of the non-exposed cohort	*	*	*	*
Ascertainment of the exposure	*	*	*	*
Demonstration that the outcome of interest was not present at start of the study	*	*	*	*
Comparability	**	**	-	**
Assessment of outcome	-	-	-	-
Was follow up long enough? (at least 6 months)	*	*	*	*
Adequacy of follow up (≥90% at 6 months)	*	*	*	*
Total score	8	8	6	8

Supplemental Table S2. Quality assessment of the two studies based on the QUALSYST guidelines.[16] Total possible scoring was out of 20, with the summary score reflecting the percent out of 20 points achieved. Yes = 2, Partially = 1, and No = 0. Possible quality assessments included: poor, fair, and good. Above a 0.65 average summary score qualified as a fair.

Study, Year (Ref #)	Huang et al., 2020[7]	Zhang et al., 2020[15]	Vijayaraman et al., 2020[16]	Li et al., 2020[21]
Question or objective sufficiently described	Yes	Yes	Yes	Yes
Study design evident and appropriate	Yes	Yes	Yes	Yes
Context for the study clear	Yes	Yes	Yes	Yes
Connection to a theoretical framework or wider body of knowledge	Yes	Yes	Yes	Yes
Sampling strategy described, relevant & justified	Partial	Partial	Partial	Partial
Data collection methods clearly described & systematic	Yes	Yes	Partial	Yes
Data analysis clearly described & systematic	Yes	Yes	Yes	Yes
Use of verification procedure(s) to establish credibility	Yes	Yes	Yes	Yes
Conclusions supported by the results	Yes	Yes	Yes	Yes
Reflexivity of the account	Yes	Yes	Yes	Yes
Total score	0.95	0.95	0.90	0.95

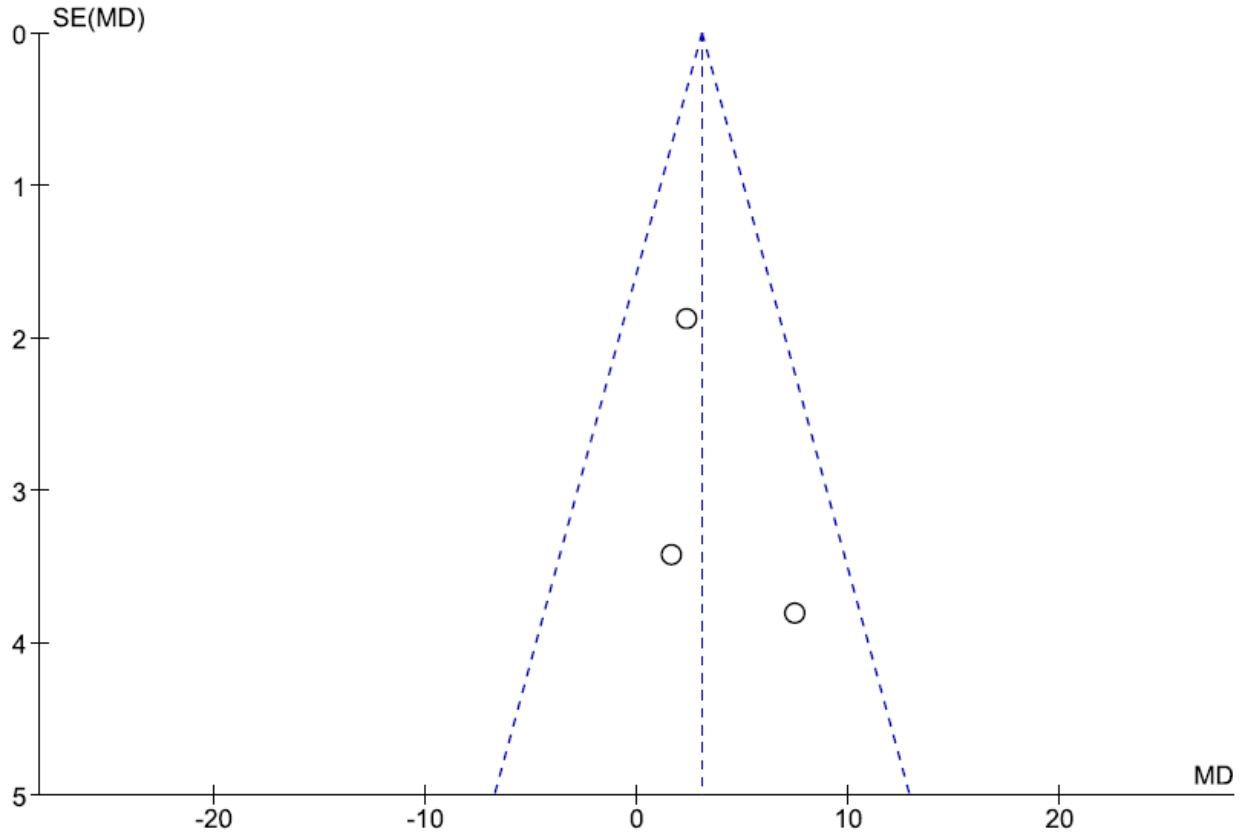
Supplemental Figure S1. Funnel Plots with 95% Confidence Interval to Evaluate the Evidence of Publication Bias

A.



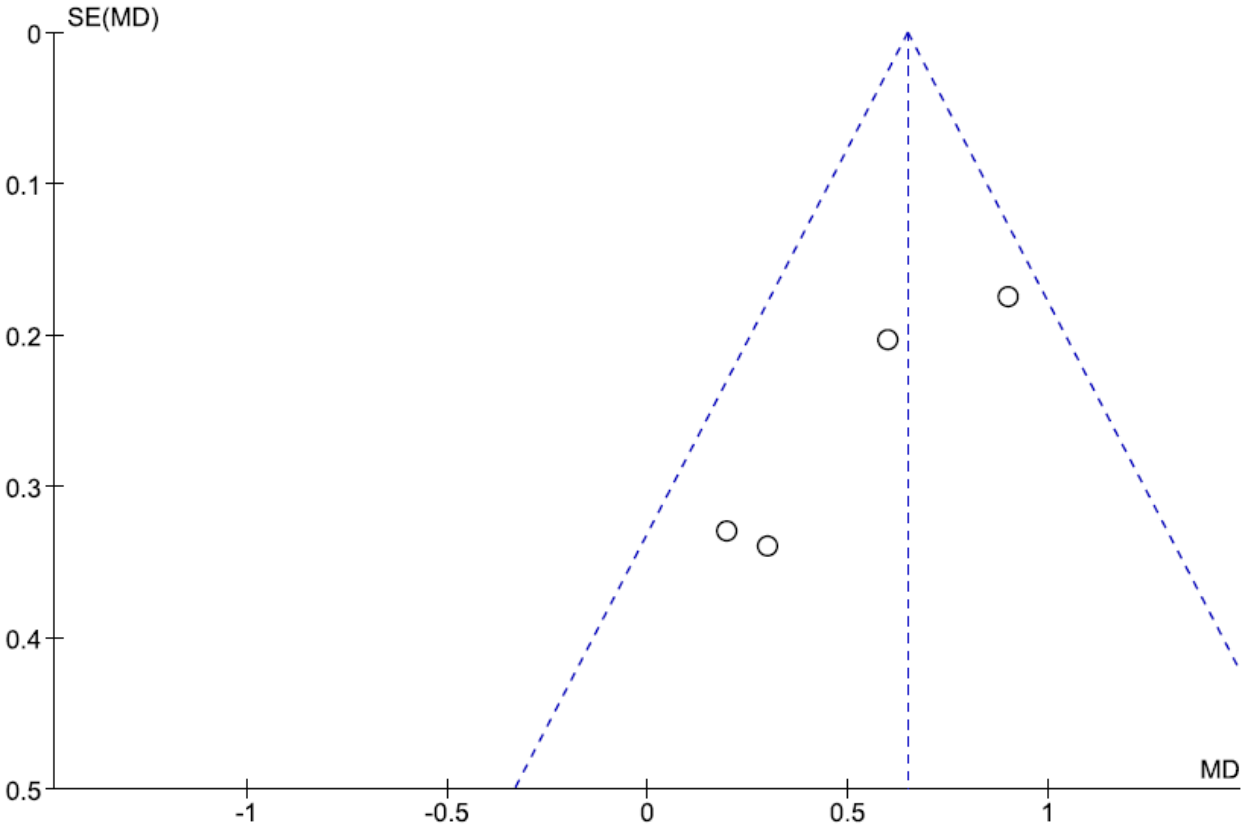
(A) Reduction in QRS duration during pacing. MD = mean difference.

B.



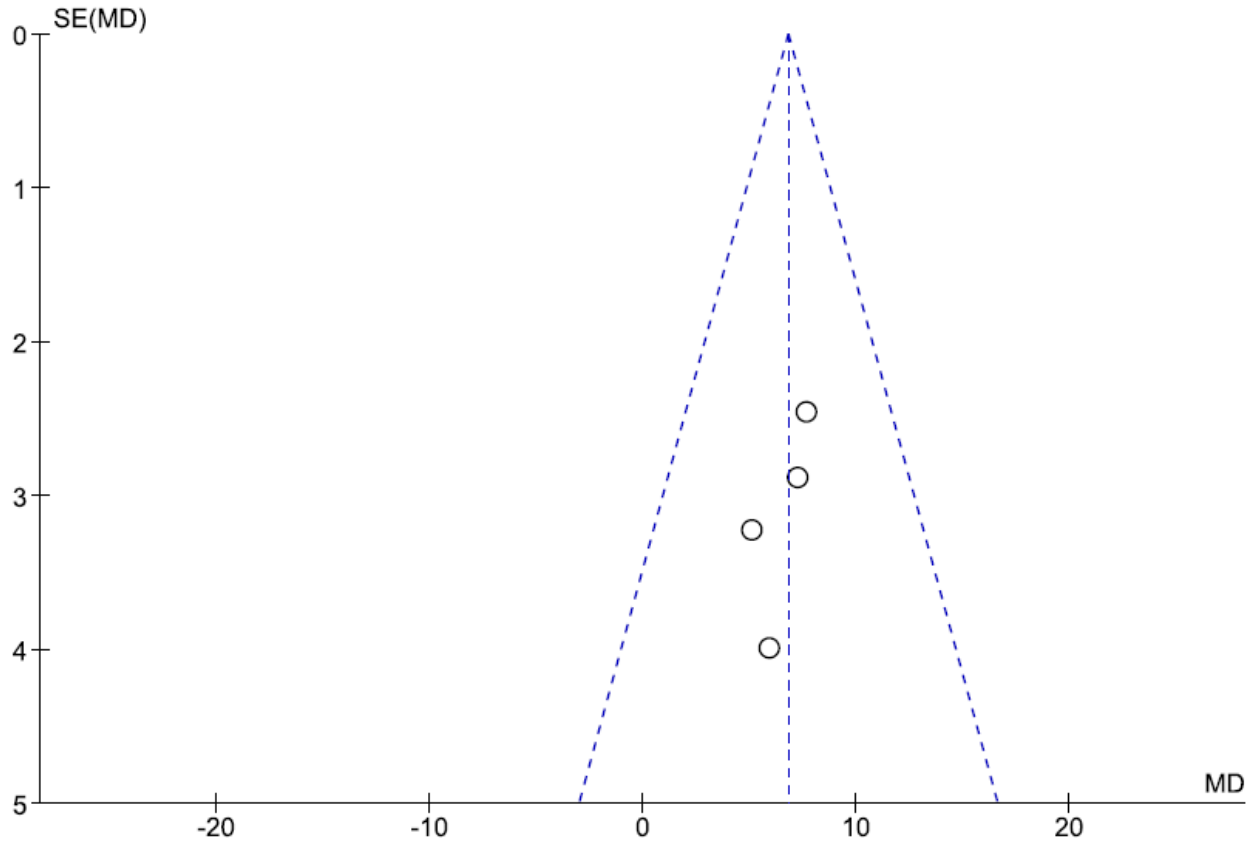
(B) Reduction in LVEDD. MD = mean difference.

C.



(C) New York Heart Association classification improvement. MD = mean difference.

D.



(D) Left ventricular ejection fraction improvement. MD = mean difference.